ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** fat-rich food	Bile aids fat digestion.
2	**(c)** 1,15,200 times	80 x 60 x 24.
3	**(d)** to provide a large surface area for absorption	Large area.
4	**(d)** digestion of all three food types	Pancreatic juice is important.
5	**(a)** 8000 mL	16 x 500.
6	**(a)** plant food takes longer to digest	Cellulose.
7	**(c)** poor protein digestion and more germs reaching the intestine	HCl needed.
8	**(c)** 480 litres	8 L x 60.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The enzyme in saliva starts acting on starch in the mouth, so R explains A.
10	**(c)** A is true but R is false.	Bile helps in fat digestion, so A is true; but bile contains no enzymes (it only breaks fat into tiny droplets), so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Respiration releases energy as heat (1); Metabolism increases after a meal (1).	1+1
12	Right sends blood to lungs nearby (1); Left pumps to the whole body (1).	1+1
13	The air in the alveoli has more oxygen than the blood (1); Oxygen moves into the blood (1).	1+1
14	Muscles need more oxygen (1); Blood must carry it faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fat digestion poor (1); Fewer enzymes (1); Special diet needed (1).	1+1+1
16	Move to site (1); Engulf germs (1); Defence (1).	1+1+1
17	Adds bulk (1); Speeds movement (1); Prevents constipation (1).	1+1+1
	OR: Muscles need more oxygen (1); Heart pumps more (1); Blood removes CO2 faster (1).	1+1+1
18	Insulin is low (1); Blood sugar rises (1); Control diet (1).	1+1+1
	OR: Frog uses oxygen (1); Needs surface air (1); Becomes distressed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) In the mouth (1). (ii) Salivary amylase (1). (iii) It churns the food and digests proteins with acid and pepsin (1). (iv) The long length gives more time for digestion and absorption (1), and villi increase the surface area for absorption of digested food into the blood (1).	1+1+1+2
20	Heart pumps blood into arteries (1); Arteries carry it to organs (1); Capillaries exchange materials with cells (1); Veins bring blood back (1); Blood returns to the heart (1).	1+1+1+1+1
	OR: Human: lungs, air (1); Fish: gills, water (1); Insect: tracheae (1); Efficiency differs (1); Adaptation to habitat (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bile contains enzymes; bile only helps in digesting **fats** and is made by the liver.
- Writing that arteries carry blood to the heart and veins from the heart; it is exactly the **opposite**.
- Confusing ingestion, digestion, absorption and egestion, especially calling egestion 'excretion'.
- Mixing up breathing (taking air in and out) with respiration (release of energy from food in cells).
- Writing that the large intestine digests food; it mainly **absorbs water**.
- In the breathing movement, saying the diaphragm moves up during inhalation; it moves **down**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 million	5 x 2.
2	**(c)** the iron-containing haemoglobin	Iron.
3	**(c)** inhalation is affected	Breathing needs it.
4	**(d)** 10.5 L	1.5 x 7.
5	**(c)** arteries have thick walls to bear high pressure	Pressure.
6	**(d)** thin air has less oxygen	Compensation.
7	**(d)** 240 hours	8 x 30.
8	**(d)** carbon dioxide	Gas exchange.
9	**(d)** A is false but R is true.	Arteries carry blood away from the heart, so A is false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but plasma being liquid does not explain oxygen transport.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Millions of alveoli (1); Large surface for gas exchange (1).	1+1
12	Adds bulk (1); Prevents constipation (1).	1+1
13	Fats would not digest properly (1); Poor absorption of fatty acids (1).	1+1
14	Dilutes urine (1); Helps pass stones (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sugar is digested quickly (1); Blood sugar drops (1); Lack of fibre and protein (1).	1+1+1
16	Waves of muscle contraction (1); Moves food along (1); Works in any posture (1).	1+1+1
17	Respiration gives heat (1); Sweat evaporates (1); Blood vessels adjust (1).	1+1+1
	OR: Dehydration (1); Concentrated wastes (1); Drink more water (1).	1+1+1
18	Training (1); Stronger muscles (1); More oxygen use (1).	1+1+1
	OR: It is easy to digest (1); It has all the nutrients (1); It protects the baby from infections (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The heart (1). (ii) Haemoglobin (1). (iii) In the alveoli (1). (iv) Muscles need more energy, so more oxygen is needed for respiration (1); faster breathing takes in more oxygen and faster heartbeat carries it to the muscles quickly (1).	1+1+1+2
20	Digestive supplies glucose (1); Respiratory supplies oxygen (1); Circulatory transports both (1); Cells release energy (1); Wastes removed (1).	1+1+1+1+1
	OR: Press to stop flow (1); Platelets clot (1); Clean wound (1); Cover (1); See a doctor (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bile contains enzymes; bile only helps in digesting **fats** and is made by the liver.
- Writing that arteries carry blood to the heart and veins from the heart; it is exactly the **opposite**.
- Confusing ingestion, digestion, absorption and egestion, especially calling egestion 'excretion'.
- Mixing up breathing (taking air in and out) with respiration (release of energy from food in cells).
- Writing that the large intestine digests food; it mainly **absorbs water**.
- In the breathing movement, saying the diaphragm moves up during inhalation; it moves **down**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** carbon dioxide	Gas exchange in tissues.
2	**(d)** partial breakdown of glucose without oxygen forms lactic acid	Anaerobic respiration.
3	**(d)** beats less often but pumps more per beat	Efficient.
4	**(d)** kidney	Made from proteins.
5	**(d)** heart, aorta, arteries, capillaries, veins, heart	Correct circuit.
6	**(a)** aorta, arteries, capillaries	Decreasing size.
7	**(a)** water, urea and salts	Waste.
8	**(a)** urea	Artificial kidney.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Muscles need more energy, so more oxygen is needed, which explains the faster breathing.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Villi give a large surface for absorption of food, so R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Damages the lining of the food pipe and stomach (1); Causes acidity (1).	1+1
12	Germs and dirt are removed (1); Stomach infections are avoided (1).	1+1
13	Cells need constant supply (1); Wastes must be removed (1).	1+1
14	Aerobic uses oxygen (1); Anaerobic occurs without oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	More oxygen needed (1); Carbon dioxide removed (1); Larger volume per breath (1).	1+1+1
16	It is the number of throbs of an artery per minute (1); It equals the heartbeat rate (1); It rises with exercise (1).	1+1+1
17	Less oxygen carrying (1); Muscles need more (1); Breathing speeds up (1).	1+1+1
	OR: Stomach: acid and pepsin act on proteins (1); Small intestine: bile and pancreatic juice act on all foods (1); Digestion is completed in the small intestine (1).	1+1+1
18	One-cell thick wall (1); Easy exchange (1); Close to every cell (1).	1+1+1
	OR: Germs (1); Cause diarrhoea (1); Contaminated water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The kidney (1). (ii) Urea (1). (iii) Two (1). (iv) Wastes like urea would collect in the blood and harm the body (1); the person would need dialysis or a kidney transplant to remove them (1).	1+1+1+2
20	Tiredness (1); Infections (1); Iron-rich diet (1); Medical advice (1); Rest (1).	1+1+1+1+1
	OR: Food has chemical energy (1); Digestion gives glucose (1); Respiration releases energy (1); Used for movement (1); Excess stored as fat (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bile contains enzymes; bile only helps in digesting **fats** and is made by the liver.
- Writing that arteries carry blood to the heart and veins from the heart; it is exactly the **opposite**.
- Confusing ingestion, digestion, absorption and egestion, especially calling egestion 'excretion'.
- Mixing up breathing (taking air in and out) with respiration (release of energy from food in cells).
- Writing that the large intestine digests food; it mainly **absorbs water**.
- In the breathing movement, saying the diaphragm moves up during inhalation; it moves **down**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND